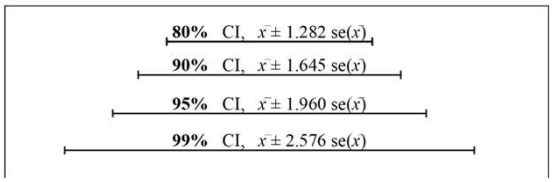
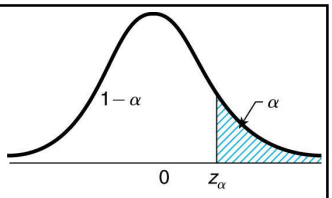
Agenda

- Last week
  - Is $\overline{\mathrm{X}} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$?
    - Large N, and Replacing σ by se$(\bar{x})$
  - χ^2_n distribution, The S^2 statistic
  - Independence of S^2 and $\bar{X}$
  - The t-distribution
  - $\mathbb{E}(S^2) = \sigma^2$
  - Confidence Interval
- Today
  - Hypothesis Testing

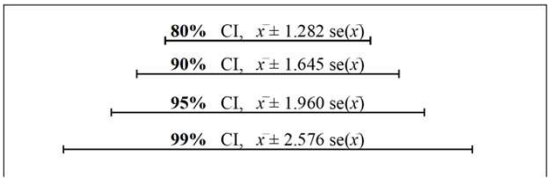


1

Context



- Given samples, report as accurate as possible a statistic
  - Make some inferences, but …
  - No measurement is accurate, so report error bounds



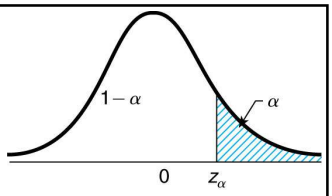
2

Recap for sample means

- With normal distributions, $(\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$
- The standard error of the sample mean is $\text{se}(\bar{x}) = \frac{s_x}{\sqrt{n}}$
- 95% confidence interval: A type of interval that contains the true value of a parameter for 95% of samples
  - For small values of n, and with a normal distribution, and with $\alpha = 0.05$, $\bar{x} \pm t_{1-\alpha/2}(n-1)\text{se}(\bar{x})$ indicates the confidence interval
- Practice: We use the same expression for confidence interval for
  - Small n, and with approximately normal distribution,
  - With the CLT approximation (large n) as a conservative practice

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Recap for sample proportions



- Small n: Cannot use the normal approximation for the Bin(n, p)
- Large n: $\hat{P} \approx \mathcal{N}(p, \frac{p(1-p)}{n})$ either via Stirling approximation or CLT applied to the Bernoulli
  - More formal: For $p \in (0,1)$ as $n \to \infty$, $\frac{\sqrt{n}(\hat{P}-p)}{\sqrt{p(1-p)}} \xrightarrow{d} \mathcal{N}(0,1)$
  - Large N: $np(1-p) > 10$ we use $\frac{(\hat{P}-p)}{\text{se}(\hat{p})} \xrightarrow{d} \mathcal{N}(0,1)$ where the standard error of the sample proportion is $\text{se}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
  - 95% confidence interval: With $\alpha = 0.05$, $\hat{p} \pm z_{1-\alpha/2}\text{se}(\hat{p})$
    - t-distribution not applicable; unlike the mean case, the variance is dependent on mean

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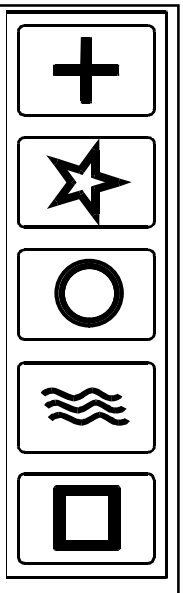
Hypothesis Testing

- Given samples, report as accurate as possible a statistic
  - Make some inferences, but …
  - No measurement is accurate, so report error bounds
  - We cannot conclusively prove a hypothesis
- Science is not a library of truth, it's a debugging process
  - Example: Mass is not absolute (Rest mass vs mass observed): $m_{\text{rel}} = \frac{m}{\sqrt{1-\frac{v^2}{c^2}}}$.
  - Being wrong is not a bug, it's a feature (engine?) of science

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Example #1

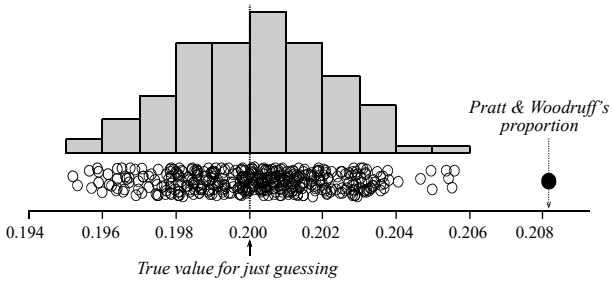
- In 1938, scientists showed 60,000 pictures
  - Students predict what is behind the card
  - If students guessed randomly there would be 12000 correct answers (proportion is 0.2)
  - … but actual result was 12489 (proportion is 0.208)
  - Is 20.8% being larger than 20% a "chance event" or
  - Hypothesis: Students can mind read correct answer in an MCQ



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Disproving the hypothesis

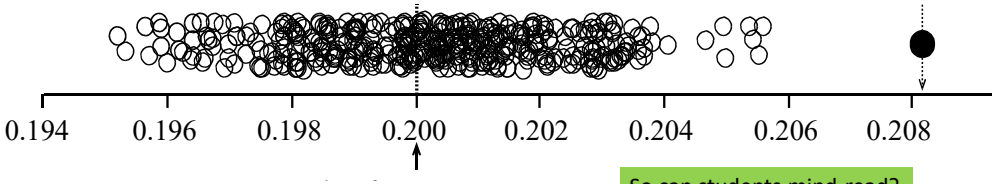
- Conduct computer simulation
  - of 400 experiments
  - Generate 60,000 possible "answers" on the back of the card
  - Guess with 0.2% probability one of the 5 cards



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Hypothesis Testing

- We cannot "rule-in" a hypothesized value for a parameter
  - For example, no amount of computer experiments can conclusively say that the proportion is 0.201 vs 0.199
  - The graph would just shift to the right (or left)
- We can only determine whether there is evidence to rule out a hypothesized value

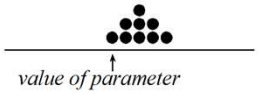


So can students mind-read?

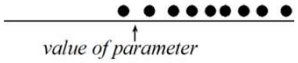
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Example #2

- In 1798, Cavendish measured the density of earth
  - After 6 experiments, he had to change his equipment resulting in the data 5.36, 5.29, 5.58, 5.65, 5.57, 5.53, 5.62, 5.29, 5.44, 5.34, 5.79, 5.10, 5.27, 5.39, 5.42, 5.47, 5.63, 5.34, 5.46, 5.30, 5.75, 5.68, 5.85
  - n = 23, sample mean = 5.483, sample sd = 0.1904 (earlier, n=29)
    - Current accepted value 5.517
  - Hypothesis: Cavendish's experiments were biased



(c) Biased, high precision

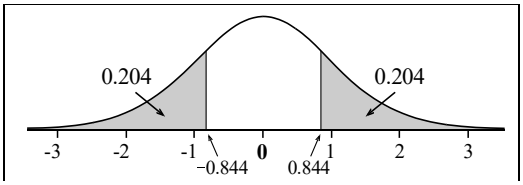


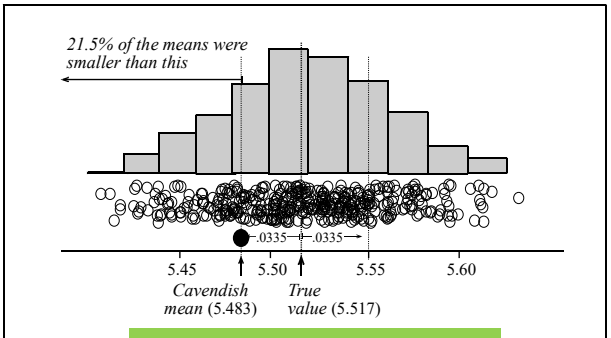
(d) Biased, low precision

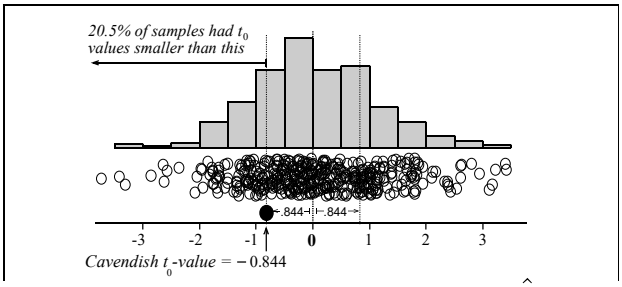
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Disproving the hypothesis

- Computer simulation
  - 400 samples from an unbiased experiment







Was Cavendish experiment biased?

$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

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Review

- What intuitive criterion do we use to determine whether the hypothesized parameter value (p=0.2 in the ESP example, and μ = 5.517 was Cavendish) was *credible* in the light of the data?
- Why was it that μ = 5.517 may be considered credible but p=0.2 was *not credible*?
- Why do we switch to measuring distance between estimate and hypothesized parameter value in terms of standard errors?
  - What do t_0-values tell us?

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Hypothesis Testing

- In settling a research hypothesis, we turn skeptical and assume the research hypothesis is false
  - The null hypothesis is H_0 (or H_o)
  - The alternate is the research hypothesis H_1 or H_a
- Is there data to reject the null hypothesis?
  - Solution:
    - Compute the test statistic
      - Is this statistic "too far" from the null hypothesis?
    - If true, then reject the null hypothesis
    - If false, cannot reject the null hypothesis

$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

Our goal is to reject the null hypothesis

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The t-test

- Using $\hat{\theta}$ to test $H_0: \theta = \theta_0$ versus H_a
  1. Calculate the test statistic t_0
  2. Calculate the p-value
  3. Interpret the p-value

P	%	Translation
> 0.12	-12%	No evidence against H0
0.1	-10%	Weak evidence against H0
0.05	-5%	Some evidence against H0
0.01	-1%	Strong evidence against H0
0.001	-0.10%	Very Strong evidence against H0

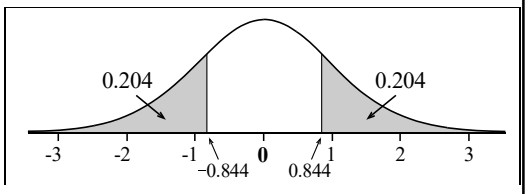
$$t_o = \frac{\hat{\theta} - \theta_0}{\mathrm{se}(\hat{\theta})}$$

Alternative hypothesis	Evidence against H_0: $\theta > \theta_0$ provided by	P-value
H_1: $\theta > \theta_0$	too much bigger than θ_0 (i.e., $\hat{\Theta} - \theta_0$ too large)	P = pr(T ≥ t_0)
H_1: $\theta < \theta_0$	too much smaller than θ_0 (i.e., $\hat{\Theta} - \theta_0$ too negative)	P = pr(T ≤ t_0)
H_1: $\theta \neq \theta_0$	too far from θ_0 (i.e., $\lvert\hat{\Theta} - \theta_0\rvert$ too large)	P = 2 pr(T ≥ $\lvert t_0 \rvert$)
		where T ~ Student(df)

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Example #2

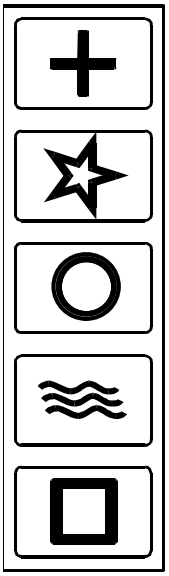
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  - After 6 experiments, he had to change his equipment resulting in the data 5.36, 5.29, 5.58, 5.65, 5.57, 5.53, 5.62, 5.29, 5.44, 5.34, 5.79, 5.10, 5.27, 5.39, 5.42, 5.47, 5.63, 5.34, 5.46, 5.30, 5.75, 5.68, 5.85
  - n = 23, sample mean = 5.483, sample sd = 0.1904 (Accepted value 5.517)
  - Research hypothesis: His last 23 experiments resulted in biased measurement
    - Null hypothesis: H_0: His experiments were not biased
    - Compute the t-statistic: -0.844
    - Compute the p-value: 0.204
    - Interpretation: Cannot refute H_o



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Example #1

- In 1938, scientists collected 60,000 pieces of data
  - If students guessed randomly
    - there would be 12000 correct answers, but actual result was 12489
  - Research hypothesis: Students mind read correct answer in an MCQ
  - Null hypothesis: proportion is 0.2
    - Compute the t-statistic: $\hat{p} = 0.20815$, p = 0.2, se=0.001657, t=4.92
    - Compute the p-value: < 10^{-6}
    - Interpret: Can refute the null hypothesis
    - Inference: Not pure guessing (maybe mind read?)



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